Example: Write a "HELLO WORLD" Program and sent it to the REMOTE

1. 在Github中建立一个远端仓库

点击右上角的"**+**",选择 new repository

先把REMOTE PULL下来，再加东西push上去

[活动摘要(简短，要干什么，add?modify?)] New Project

写内容1. 2.

！ AR 填写开发需求的序号

.md格式 markdone

git push 仓库名 本地branch:远程仓库branch

.gitignore 文件中输入相关

2. 在Ubuntu中创建python文件

1. 打开文件准备存放的文件夹
2. Open in terminal
3. 输入 nano hello.py
   a. nano:终端文本编辑器， 用来 **新建/打开** 某个文件 写/改 代码
   b. **.py** 结尾不会被自动识别是python代码，只是方便看，**真正起到作用的**是后面运行代码时的 **python3**
4. 进入python,输入需要的代码， ctrl +o 保存， enter, ctrl+x 退出
5. 出来后第一时间检查文件格式是不是python (第一次完成时文件格式输出成了 hello.py.save)
   a. 修改文件名回到python格式： `mv hello.py.save hello.py`
6. 运行代码：open in terminal, `python3 <文件名>` ---->得到运行结果
   a. python3 是python的解释器程序， “3” 是版本号
7. 此时使用一次 `git status` 记录状态:

```
duan@duan-virtual-machine:~/桌面/git_test$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        hello.py

nothing added to commit but untracked files present (use "git add" to track)
duan@duan-virtual-machine:~/桌面/git_test$
```

a. No commit

b. No tracked, 并且给出第一步建议

8. 使用 `git add hello.py` 存储到staging aera中并记录状态

```
duan@duan-virtual-machine:~/桌面/git_test$ git add hello.
py
duan@duan-virtual-machine:~/桌面/git_test$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   hello.py

duan@duan-virtual-machine:~/桌面/git_test$
```

9. 使用 `git remote add <shortname> <url>` 与在Github中创建的Remote相连并给他一个名为"T1"的简写； 对是否连接成功进行验证 `git remote -v`

```
duan@duan-virtual-machine:~/桌面/git_test$ git remote add T1 https://github.com
/pujipia/Test_1
duan@duan-virtual-machine:~/桌面/git_test$ git remote
T1
duan@duan-virtual-machine:~/桌面/git_test$ git remote -v
T1      https://github.com/pujipia/Test_1 (fetch)
T1      https://github.com/pujipia/Test_1 (push)
```

10. Commit 写好的 hello.py 文件，并输出对应的解释， **命令行中不包含要commit的文件名！**

a. Commit 只管提交所有在staging area中的全部文件

b. 有不想提交的在staging area 中的文件： `git restore --staged <file_name>`

```
duan@duan-virtual-machine:~/桌面/git_test$ git commit -m "Add hello.py, the fir
st GIt test"
[master (root-commit) 14895b6] Add hello.py, the first GIt test
 1 file changed, 1 insertion(+)
 create mode 100644 hello.py
```

c. commit后出现了这样的界面：

```
duan@duan-virtual-machine: ~/...
  /home/duan/桌面/Test_1/.git/COMMIT_EDITMSG   Modified
this is a test for upload a file

# Please enter the commit message for your changes. Lines>
# with '#' will be ignored, and an empty message aborts t>
#
# On branch master
# Your branch is up-to-date with 'origin/master'.
#
# Changes to be committed:
#       new file:   MC_SiBulk_Mobility_fixed/Include/Si-b>
#       new file:   MC_SiBulk_Mobility_fixed/Include/Si-f>
#       new file:   MC_SiBulk_Mobility_fixed/Include/Si-l>
#       new file:   MC_SiBulk_Mobility_fixed/Include/Si-l>
#       new file:   MC_SiBulk_Mobility_fixed/Include/Si-l>
#       new file:   MC_SiBulk_Mobility_fixed/Include/Si-o>
#       new file:   MC_SiBulk_Mobility_fixed/Include/Si.d>
#       new file:   MC_SiBulk_Mobility_fixed/Include/SiGe>
#       new file:   MC_SiBulk_Mobility_fixed/Include/Strn>
#       new file:   MC_SiBulk_Mobility_fixed/Include/cons>
File to insert [from ./]:
^G Get Help        M-F New Buffer      ^X Execute Command
^C Cancel          M-N No Conversion   ^T To Files
```

11. 把本地commit的文件 "推 push"到网上去 `git push <remote> <branch>`
    a. 我**忘记我的REMOTE 的 shortname**怎么办？ `git remote`
    b. 我犯的错误：
        i. 第一次： 不是origin, command 中的 <remote> 是**远端仓库名**，在对接时被命名为 T1
        ii. 第二次： 不是main, 我的本地分支名是 "master" 可以用 `git branch` 来查询本地分支名
        iii. 代码中的 `-u` 代表什么：

        执行完： `git push -u origin master`

        Git 就记住了：我现在这个本地分支 `master`，默认对应远程的 `origin/master`

        这样以后你就不用每次都写完整的：

        `git push origin master` | `git pull origin master`

        而可以直接写：

        `git push` | `git pull`

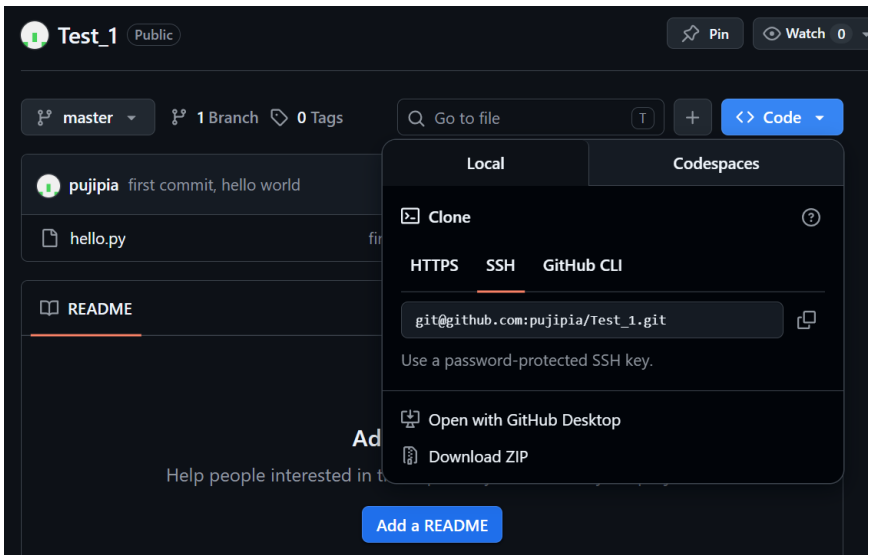
```
duan@duan-virtual-machine:~/桌面/git_test$ git push -u origin main
error: src refspec main does not match any
error: failed to push some refs to 'origin'
duan@duan-virtual-machine:~/桌面/git_test$ git push -u T1 main
error: src refspec main does not match any
error: failed to push some refs to 'https://github.com/pujipia/Test_1'
duan@duan-virtual-machine:~/桌面/git_test$ git push -u T1 master
```

        iv. 该问题主要出现在网络问题

```
duan@duan-virtual-machine:~/桌面/git_test$ git push -u T1 master
fatal: unable to access 'https://github.com/pujipia/Test_1/': GnuTLS recv error
 (-54): Error in the pull function.
```

v. 把Github中的 "HTTPS" 改成 "SSH" 有概率解决部分问题：

   1. 点击进入"Repository"
   2. 点击要进入的仓库名进入对应仓库
   3. 点击"code"



vi. `git reflog` **:记录 `HEAD` 和分支引用“动过哪里”**，也就是帮你找回“我刚才切到哪了、reset 到哪了、commit 之前在哪”

vii. cp ../hello.py . cp代表copy, ../hello .,前两个点代表从前面的前面(桌面) 把 hello.py 拷到一个点（Test1）中

```
duan@duan-virtual-machine:~/桌面/Test_1$ git ../hello.py .
```

viii. Sudo apt: sudo means supervisor do

3. 尝试修改文件并通过Git恢复旧版本

1. 修改毕设文件MAIN中的option=6,save
2. `git status` --准确告诉我我修改了哪个地方

```
duan@duan-virtual-machine:~/桌面/Test_1$ git status
On branch master
Your branch is up-to-date with 'origin/master'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committe
d)
  (use "git restore <file>..." to discard changes in worki
ng directory)
        modified:   MC_SiBulk_Mobility_fixed/main.f

no changes added to commit (use "git add" and/or "git comm
it -a")
```

3. 如果我再修改一个别的文件会发生什么？我删掉了run的数据 driftSi.dat --新增一个**被删除文件**

```
duan@duan-virtual-machine:~/桌面/Test_1$ git status
On branch master
Your branch is up-to-date with 'origin/master'.

Changes not staged for commit:
  (use "git add/rm <file>..." to update what will be commi
tted)
  (use "git restore <file>..." to discard changes in worki
ng directory)
        deleted:    MC_SiBulk_Mobility_fixed/driftSi.dat
        modified:   MC_SiBulk_Mobility_fixed/main.f

no changes added to commit (use "git add" and/or "git comm
it -a")
```

4. 我只暂存并commit modified file 为什么会有1+ 1-：

把一行旧内容改成了一行新内容。

Git 在统计时会默认为删除一行旧的、添加一行新的

```
duan@duan-virtual-machine:~/桌面/Test_1$ git status
On branch master
Your branch is up-to-date with 'origin/master'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   MC_SiBulk_Mobility_fixed/main.f

Changes not staged for commit:
  (use "git add/rm <file>..." to update what will be commi
tted)
  (use "git restore <file>..." to discard changes in worki
ng directory)
        deleted:    MC_SiBulk_Mobility_fixed/driftSi.dat
```

```
duan@duan-virtual-machine:~/桌面/Test_1$ git commit -m "I
change the option in main.f and let it=6"
[master 07a541b] I change the option in main.f and let it=
6
 1 file changed, 1 insertion(+), 1 deletion(-)
```

5. Github上只会显示发生了变动的地方

1 file changed +1 -1 lines changed

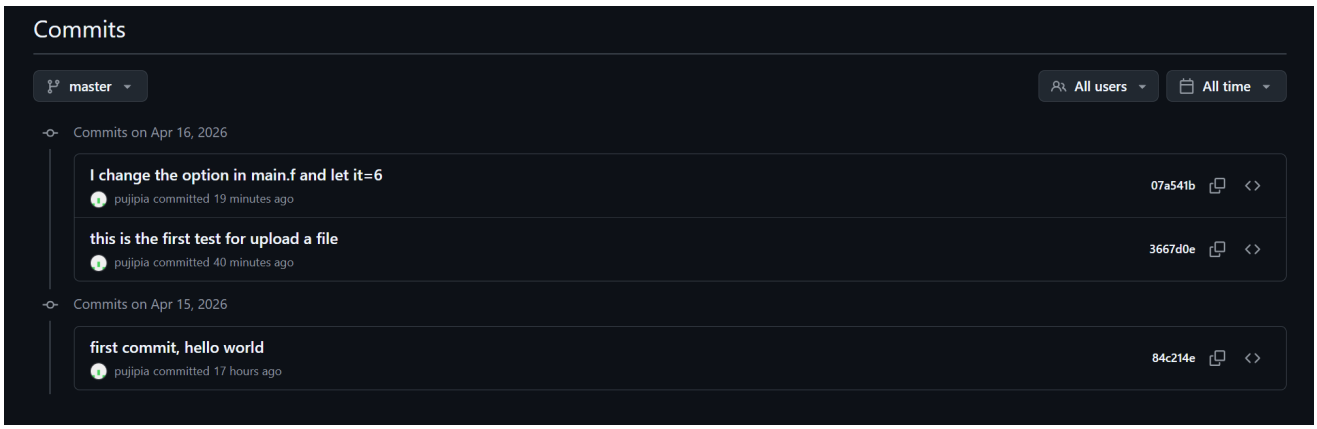
MC_SiBulk_Mobility_fixed/main.f

```diff
@@ -60,7 +60,7 @@
 c-- 1=AlGaAs, 2=GaP, 3=InAlAs, 4=InP, 5=AlAs, 6=InGaAs, 7=Si
 c   Only Silicon is here at the moment.
 c
-      option=7        ! <-------------
+      option=6        ! <-------------
 c
       print *,' 7=Si'
       print *,' SELECTED:'
```

6. 我如果想把本地版本恢复成初始版本：`git reset --hard <haxi值>`

```
duan@duan-virtual-machine:~/桌面/Test_1$ git reset --hard
 3667d0e
HEAD is now at 3667d0e this is the first test for upload a
 file
```

- Github中得到各版本提交的哈希：



- 此时结果：

```
duan@duan-virtual-machine:~/桌面/Test_1$ git status
On branch master
Your branch is behind 'origin/master' by 1 commit, and can
 be fast-forwarded.
  (use "git pull" to update your local branch)

nothing to commit, working tree clean
```

  - Explaination:

> **你现在本地的 `master` 比远程的 `origin/master` 落后 1 次提交。**
>
> 也就是说：
>
> - 远程仓库里有一个比你本地更新的版本
> - 你本地还没到那个版本
> - 但是这次更新很简单，可以直接“快进”到远程版本，不会有冲突

- 本地的main.f 恢复了 opion = 7:

```
c
c-- Change OPTION to the appropriate material HERE:
c-- 1=AlGaAs, 2=GaP, 3=InAlAs, 4=InP, 5=AlAs, 6=InGaAs, 7=Si
c   Only Silicon is here at the moment.
c
        option=7        ! <--------------
c
        print *,' 7=Si'
        print *,' SELECTED:'
        if (option .eq. 7) then
         include 'Include/Si.dat'
```